

CHIAYI AP, Taiwan

WMO: 467460

Lat: 23.462N

Lon: 120.393E

Elev: 85

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	47.5	50.0	41.4	38.5	50.1	44.7	43.8	52.3	21.9	63.3	19.3	63.2	5.5	10	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	93.2	82.1	91.8	81.6	91.3	81.4	83.8	90.6	82.8	89.8	82.1	89.0	10.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.2	168.0	88.7	80.9	160.9	87.6	80.4	158.4	87.1	48.0	90.6	47.0	90.1	46.2	89.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.9	16.6	14.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	35.8	86.4	3.3	1.4	33.4	87.4	31.4	88.3	29.6	89.1	27.1	90.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.7	63.2	64.0	68.0	74.7	79.5	82.8	84.1	83.7	81.6	77.3	71.6	65.4
	DBStd	8.73	4.78	5.52	6.31	5.66	3.56	2.84	2.00	2.12	2.78	3.41	4.64	4.81
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	281	90	77	44	8	0	0	0	0	0	0	8	54
	CDD50	9025	410	392	560	740	915	985	1058	1044	948	847	649	477
	CDD65	3830	35	49	138	299	450	535	593	579	498	382	207	65
	CDH74	35427	155	233	751	2105	4000	5920	6960	6524	4891	2615	1003	270
	CDH80	13069	17	36	183	663	1416	2398	2928	2592	1767	836	208	25
Wind	WSAvg	6.0	7.1	7.0	6.8	5.7	5.2	6.2	5.8	5.8	5.1	5.3	5.8	6.1
Precipitation	PrecAvg	77.4	1.1	2.0	2.2	4.3	6.8	14.5	16.5	17.2	8.8	1.9	1.1	0.9
	PrecMax	127.2	3.5	12.9	6.3	18.0	16.0	36.1	37.8	46.0	40.4	12.9	8.3	3.7
	PrecMin	40.5	0.0	0.0	0.2	0.0	0.4	0.9	2.4	3.9	1.6	0.0	0.0	0.0
	PrecStd	22.2	0.9	2.8	1.9	4.1	3.9	8.9	9.4	12.0	9.1	2.9	1.8	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.4	84.2	88.0	91.5	92.9	94.8	94.0	93.5	92.5	89.9	86.8	82.8
		MCWB	71.9	73.2	76.2	79.5	81.3	82.4	82.1	82.6	81.6	79.6	76.9	73.5
	2%	DB	78.4	80.3	84.5	88.7	90.0	93.2	93.1	92.2	91.2	88.1	84.5	79.8
		MCWB	68.3	70.7	74.7	78.1	80.0	82.0	82.1	82.2	81.5	79.0	75.6	70.6
	5%	DB	75.1	76.8	82.1	86.3	89.2	91.6	91.7	91.4	89.6	86.4	82.5	77.0
		MCWB	66.2	68.8	73.4	76.9	79.7	81.4	81.6	81.9	80.8	77.9	74.3	68.7
	10%	DB	71.8	73.3	78.6	84.1	87.5	89.8	91.0	89.9	88.1	84.7	80.1	74.8
		MCWB	64.6	66.7	71.5	76.0	79.0	80.7	81.3	81.3	80.3	76.9	72.7	67.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.6	74.5	77.7	80.4	82.8	84.5	84.8	85.3	83.9	82.0	78.8	74.7
		MCDB	80.3	82.6	85.8	88.9	89.5	91.9	91.1	90.8	89.9	88.2	85.0	81.7
	2%	WB	69.4	71.7	75.4	78.9	81.4	83.0	83.5	83.7	82.6	79.9	76.8	71.7
		MCDB	76.5	78.5	83.3	87.1	88.4	90.6	90.6	90.2	89.1	86.3	83.5	78.0
	5%	WB	67.2	69.5	73.7	77.7	80.1	82.0	82.5	82.7	81.6	78.7	75.1	69.6
		MCDB	73.2	75.0	80.3	85.2	87.0	89.5	89.8	89.3	88.0	85.2	80.8	75.3
	10%	WB	65.3	67.3	72.0	76.4	79.2	81.0	81.6	81.8	80.5	77.5	73.4	67.7
		MCDB	70.7	72.7	77.5	82.9	85.5	87.9	88.6	88.1	86.7	83.7	78.3	73.0
Mean Daily Temperature Range	5% DB	MDBR	17.8	16.8	16.6	15.7	14.2	13.5	13.4	13.3	13.8	16.1	17.2	19.0
		MCDBR	21.3	20.4	21.0	18.5	16.0	15.6	15.0	14.7	14.9	16.8	18.3	21.1
		MCWBR	12.9	11.8	12.6	10.1	8.6	7.8	7.3	7.7	7.8	9.2	11.1	12.9
	5% WB	MCDBR	19.3	19.3	19.4	17.6	14.9	14.6	14.2	13.9	14.2	16.2	17.2	19.3
		MCWBR	12.3	11.9	12.1	10.2	8.5	8.0	7.6	8.0	8.0	9.6	10.9	12.6
Clear-Sky Solar Irradiance	taub	0.509	0.563	0.691	0.693	0.586	0.529	0.540	0.597	0.628	0.610	0.532	0.522	
	taud	1.969	1.866	1.620	1.665	1.938	2.109	2.072	1.900	1.803	1.796	1.973	1.955	
	Ebn at Noon	234	231	209	212	236	247	245	231	220	215	226	224	
	Edh at Noon	52	62	83	81	61	51	53	63	68	65	52	52	
All-Sky Solar Radiation	RadAvg	1073	1205	1308	1404	1526	1609	1753	1579	1539	1432	1137	1044	
	RadStd	118	150	138	174	188	168	128	164	124	89	95	103	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	+0.50	N/A	N/A	N/A	N/A	+0.32	+0.45	N/A	N/A	+218	+200

Nomenclature: See separate page